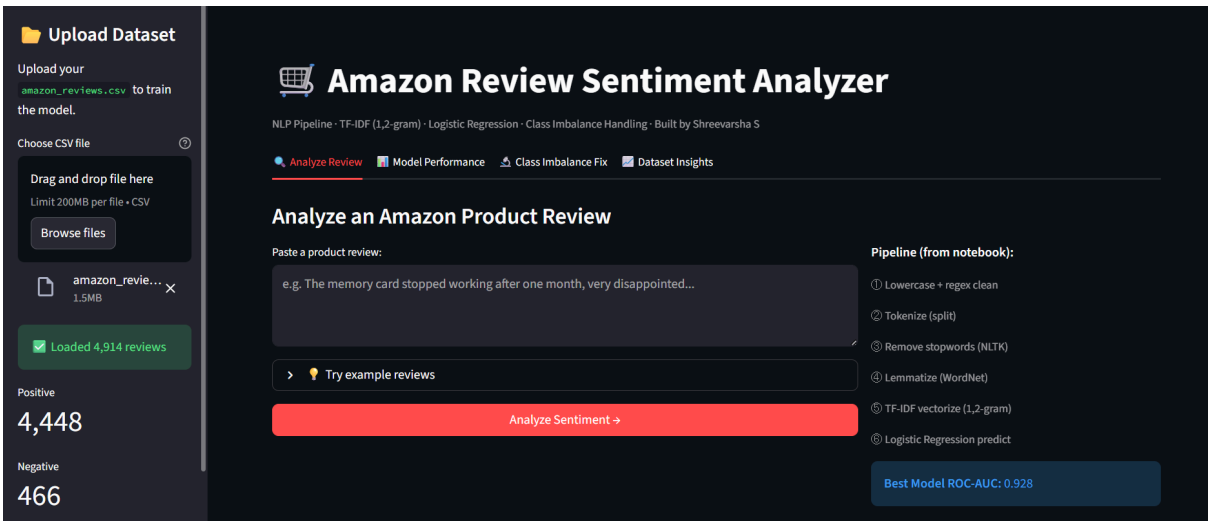
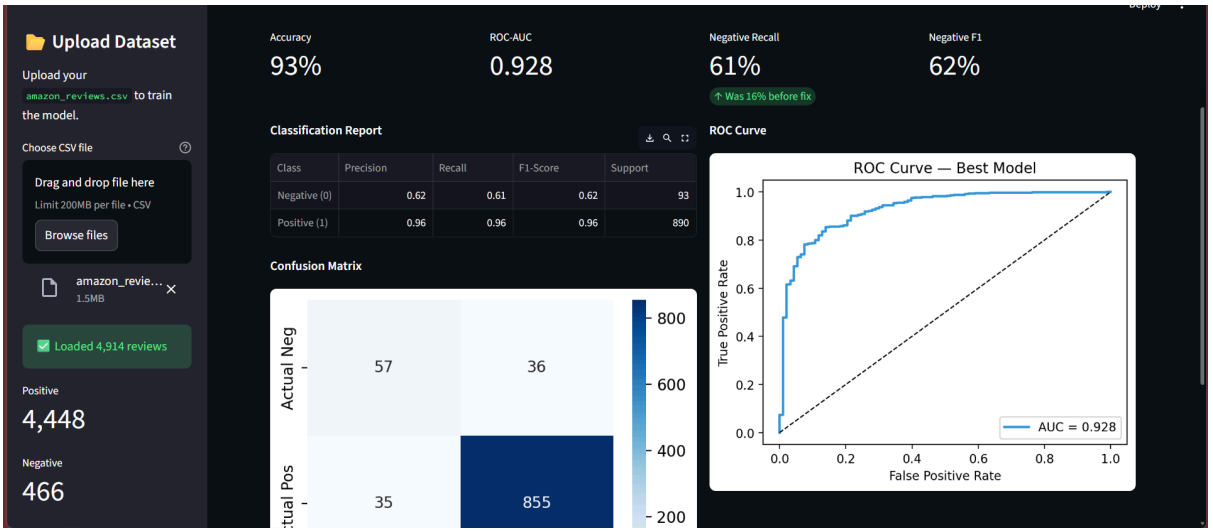

Upload Dataset
Upload your amazon_reviews.csv to train the model.
Choose CSV file
Drag and drop file here
Limit 200MB per file • CSV
Browse files
amazon_revie...
1.5MB
Loaded 4,914 reviews
Positive
4,448
Negative
466
Amazon Review Sentiment Analyzer
NLP Pipeline · TF-IDF (1,2-gram) · Logistic Regression · Class Imbalance Handling · Built by Shreevarsha S
Analyze Review
Model Performance
Class Imbalance Fix
Dataset Insights
Analyze an Amazon Product Review
Paste a product review:
e.g. The memory card stopped working after one month, very disappointed...
Try example reviews
Analyze Sentiment →
Pipeline (from notebook):
① Lowercase + regex clean
② Tokenize (split)
③ Remove stopwords (NLTK)
④ Lemmatize (WordNet)
⑤ TF-IDF vectorize (1,2-gram)
⑥ Logistic Regression predict
Best Model ROC-AUC: 0.928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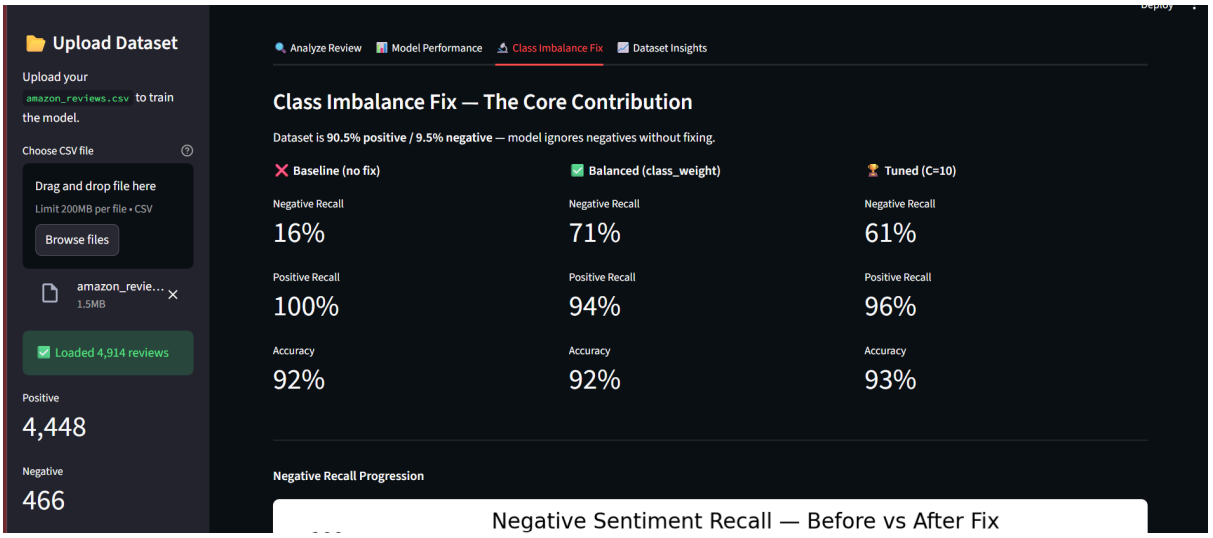
Limit 200MB per file • CSV
Browse files
amazon_revie...
1.5MB
Loaded 4,914 reviews
Positive
4,448
Negative
466
Accuracy
93%
ROC-AUC
0.928
Negative Recall
61%
↑ Was 16% before fix
Negative F1
62%
Classification Report
Class | Precision | Recall | F1-Score | Support
Negative (0) | 0.62 | 0.61 | 0.62 | 93
Positive (1) | 0.96 | 0.96 | 0.96 | 890
Confusion Matrix
Actual Neg
Actual Pos
57
36
35
855
800
600
400
200
ROC Curve
ROC Curve — Best Model
True Positive Rate
False Positive Rate
AUC = 0.928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Browse files
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1.5MB
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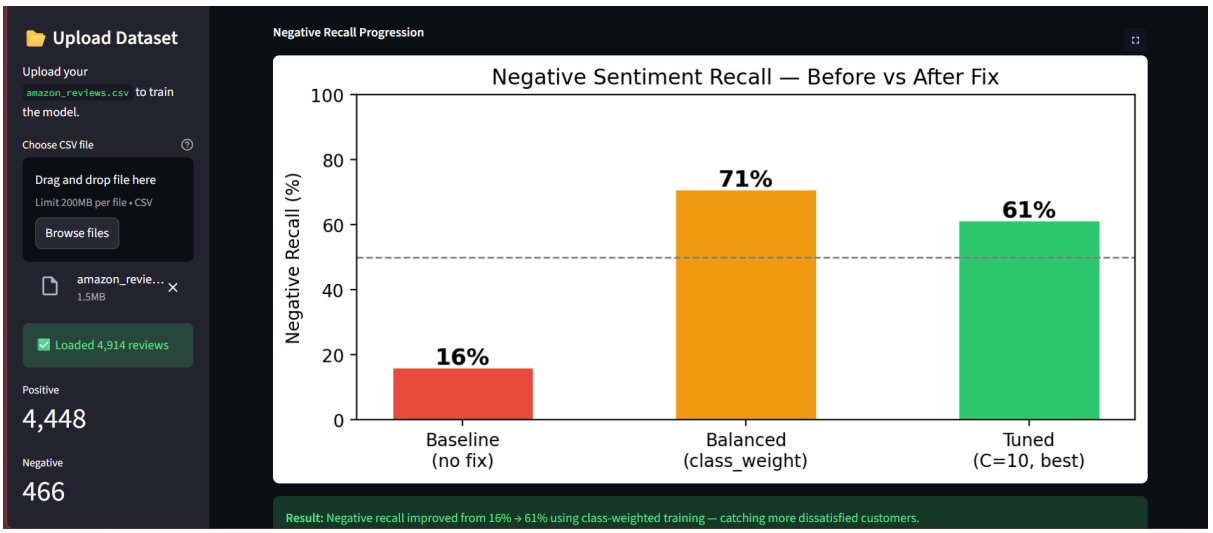
Analyze Review
Model Performance
Class Imbalance Fix
Dataset Insights
Class Imbalance Fix — The Core Contribution
Dataset is 90.5% positive / 9.5% negative — model ignores negatives without fixing.
Baseline (no fix)
Negative Recall
16%
Positive Recall
100%
Accuracy
92%
Balanced (class_weight)
Negative Recall
71%
Positive Recall
94%
Accuracy
92%
Tuned (C=10)
Negative Recall
61%
Positive Recall
96%
Accuracy
93%
Negative Recall Progression
Negative Sentiment Recall — Before vs After Fix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Limit 200MB per file • CSV
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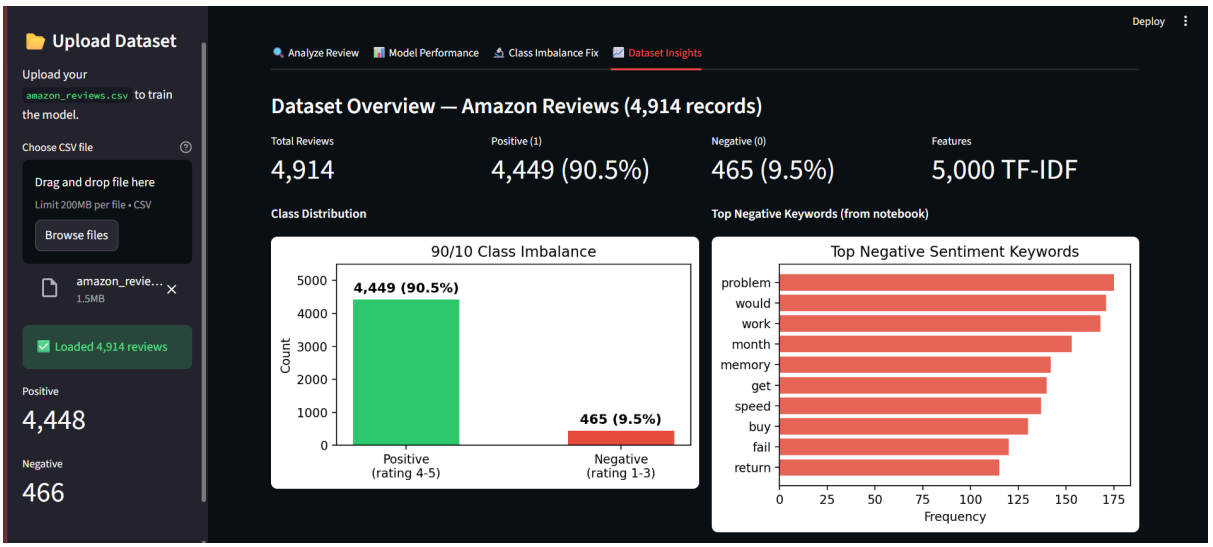
1.5MB
Loaded 4,914 reviews
Positive
4,448
Negative
466
Negative Recall Progression
Negative Sentiment Recall — Before vs After Fix
Negative Recall (%)
100
80
60
40
20
0
16%
71%
61%
Baseline (no fix)
Balanced (class_weight)
Tuned (C=10, best)
Result: Negative recall improved from 16% → 61% using class-weighted training — catching more dissatisfied customers.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Negative
466
Deploy
Analyze Review
Model Performance
Class Imbalance Fix
Dataset Insights
Dataset Overview — Amazon Reviews (4,914 records)
Total Reviews
4,914
Positive (1)
4,449 (90.5%)
Negative (0)
465 (9.5%)
Features
5,000 TF-IDF
Class Distribution
90/10 Class Imbalance
Count
5000
4000
3000
2000
1000
0
4,449 (90.5%)
465 (9.5%)
Positive (rating 4-5)
Negative (rating 1-3)
Top Negative Keywords (from notebook)
Top Negative Sentiment Keywords
problem
would
work
month
memory
get
speed
buy
fail
return
0
25
50
75
100
125
150
175
Frequency